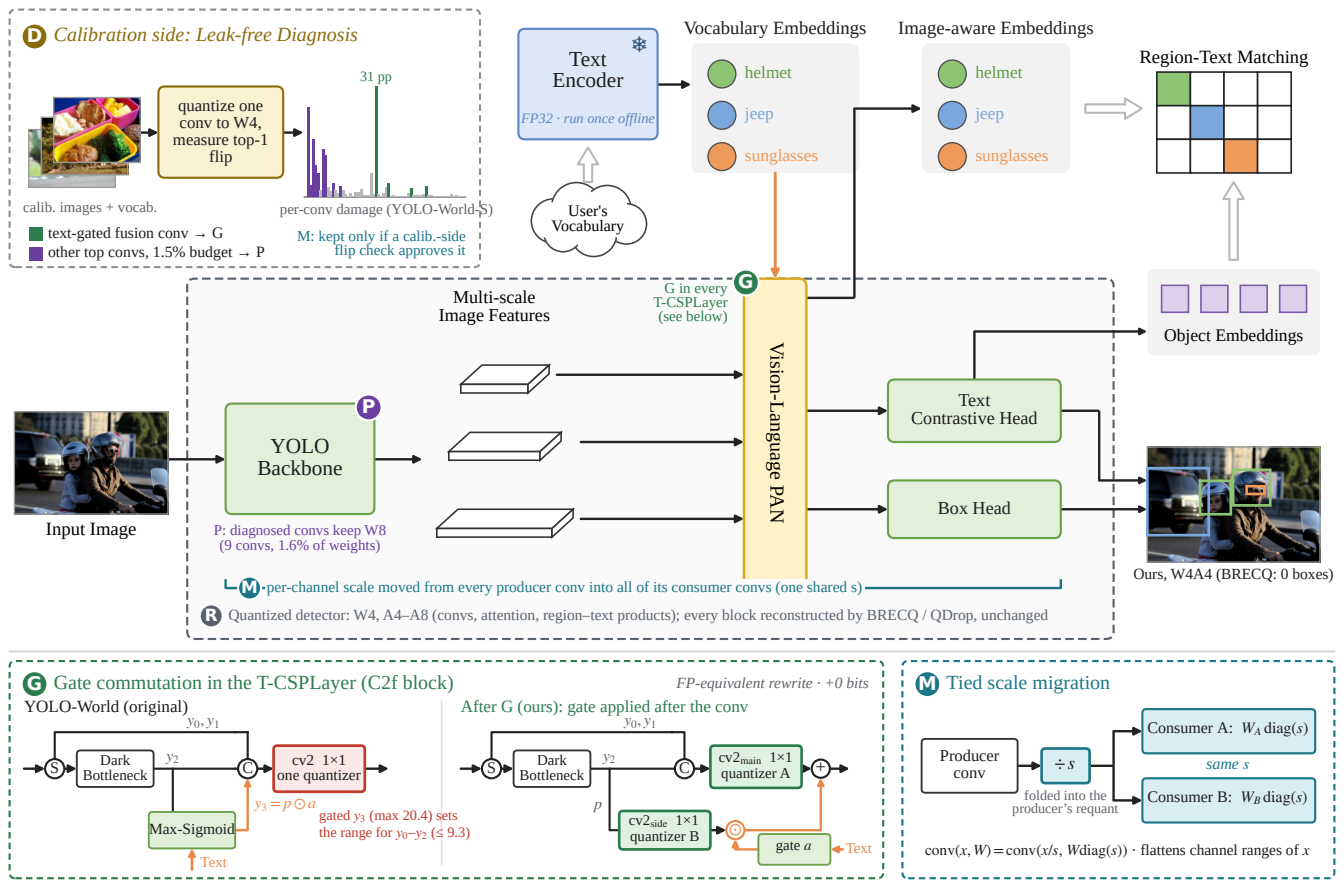

D Calibration side: Leak-free Diagnosis
quantize one conv to W4, measure top-1 flip
31 pp
calib. images + vocab.
per-conv damage (YOLO-World-S)
text-gated fusion conv → G
other top convs, 1.5% budget → P
M: kept only if a calib.-side flip check approves it
Text Encoder
FP32 · run once offline
User's Vocabulary
Vocabulary Embeddings
helmet
jeep
sunglasses
Image-aware Embeddings
Region-Text Matching
Object Embeddings
Multi-scale Image Features
G in every T-CSPLayer (see below)
Vision-Language PAN
Input Image
YOLO Backbone
P: diagnosed convs keep W8 (9 convs, 1.6% of weights)
Text Contrastive Head
Box Head
Ours, W4A4 (BRECQ: 0 boxes)
M per-channel scale moved from every producer conv into all of its consumer convs (one shared s)
R Quantized detector: W4, A4–A8 (convs, attention, region–text products); every block reconstructed by BRECQ / QDrop, unchanged
G Gate commutation in the T-CSPLayer (C2f block)
FP-equivalent rewrite · +0 bits
YOLO-World (original)
Dark Bottleneck
cv2 1×1 one quantizer
Max-Sigmoid
Text
gated y_3 (max 20.4) sets the range for y_0–y_2 (≤ 9.3)
After G (ours): gate applied after the conv
cv2main 1×1 quantizer A
cv2side 1×1 quantizer B
gate a
M Tied scale migration
Producer conv
÷ s
folded into the producer's requant
Consumer A: W_A diag(s)
same s
Consumer B: W_B diag(s)
conv(x, W) = conv(x/s, W diag(s)) · flattens channel ranges of x